

ADEOLUWA S. ADEKANMBI CURRICULUM VITAE

CONTACT

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EDUCATION

Howard University, Washington, D.C.

May 2026

Bachelor of Science in Electrical Engineering, in progress

GPA: 3.87/4.0

TECHNICAL SKILLS

Languages: Python, C++, Golang, JavaScript, VHDL

Frameworks: PyTorch, TensorFlow, Scikit-learn

Tools: LTSpice, Smath Studio, AWS EC2, Git, Docker, MATLAB, FPGA, Vivado, Raspberry Pi, ESP-32, Arduino, Jupyter, Microsoft Office Suite

RESEARCH AND WORK EXPERIENCE

MIT Summer Research Program

Summer 2025

Machine Olfaction Research Intern, Media Lab

Cambridge, MA

- Collected over 4 hours of multi-sensor time-series data on food mixtures using portable gas sensors.
- Trained ML models detecting peanuts in chocolate mixtures with ~80% confidence and ~42% rejection of non-peanut samples in real-time preliminary trials.
- Performed real-time odor classification using embedded hardware with low latency.

MIT Summer Research Program

Summer 2024

Machine Learning Research Intern, Laboratory for Information and Decision Systems

Cambridge, MA

- Developed MultiSenseNet, a framework enabling multi-task learning across visual and textual modalities.
- Designed and trained CNN, LSTM, and fusion architectures to evaluate generalization across diverse ML tasks.
- Investigated transformer scalability and transfer learning to develop foundation models for renewable energy systems.

Amazon Web Services (AWS)

Summer 2023

Software Development Engineer Intern, Annapurna Labs

Cupertino, CA

- Developed command line tool for benchmarking ML model performance, reducing runtime analysis by 400%.
- Optimized visualization of metrics for internal UI, improving comparison efficiency by 4x.
- Authored design documentation and implemented unit tests for production-level deployment.

Howard University VIP Project

Fall 2023 – Fall 2024

Student Researcher

Washington, D.C.

- Implemented a small-scale autonomous pump control system using capacitive sensors, solid-state relays, and Programmable Logic Controllers in collaboration with DC Water and Howard University.
- Built a Raspberry Pi-based radial pulse monitoring system for ADC data acquisition, and GPIO interfacing.

College of Engineering & Architecture (CEA)

Fall 2024 – Present

Drop-In Tutor

Washington, D.C.

- Tutored 30+ undergraduate students in math, physics, chemistry, and core electrical engineering courses.
- Clarified complex concepts through step-by-step problem solving and hands-on examples to improve student comprehension.
- Fostered an engaging learning environment by adapting explanations to diverse learning styles and providing real-time feedback.

LEADERSHIP EXPERIENCE

Howard University Bethel Campus Fellowship (BCF)

Fall 2023 – Present

President

Washington, D.C.

- Lead a 6-member executive board to serve and support over 60 Howard students through Bible studies, and other community-building activities.
- Organized campus outreaches impacting 200+ students, fostering spiritual growth and community engagement.
- Secured over \$11,000 in funding to enable student attendance at the BCF National Conference, promoting leadership, service, and personal development.

College of Engineering & Architecture

Fall 2023 – Spring 2024

Student Council Hackathon Coordinator

Washington, D.C.

- Coordinated Howard students' participation in hackathons, leading to multiple wins and recognition.
- Built partnerships with peer institutions to reduce financial and transportation barriers to attendance.
- Co-led the organization of Howard's inaugural CEA Hackathon, fostering innovation and technical skill development.

College of Engineering & Architecture

Fall 2022 – Spring 2023

Student Council E-Board Intern

Washington, D.C.

- Secured \$40,000 in sponsorships to fund student events and initiatives across the College.
- Collaborated with E-Board members to develop impactful student policies and programs.
- Assisted in planning and executing events serving 200+ students and faculty, strengthening community engagement.

PROJECTS

MoneyMind | [Website](#)

March 2025

- Developed MoneyMind, a web platform combining financial management tools and a student marketplace powered by AI insights.

ResQlink | [Website](#)

February 2025

- Built ResQlink, an AI-powered disaster response app that alerts users of nearby hazards, routes them to shelters, and connects rescue teams.

MindMatters | [Website](#)

March 2024

- Created MindMatters, an AI chatbot supporting first-generation students with personalized academic, financial, and career guidance.

Autonomous Pump System | [Website](#)

November 2024

- Developed a small-scale autonomous pump control system using capacitive sensors, solid-state relays, and Programmable Logic Controllers in collaboration with DC Water and Howard University.

Digital Combination Lock | [Website](#)

November 2023

- Built a Finite State Machine (FSM) for a digital combination lock with state transitions based on keypad combination entry.

POSTER PRESENTATIONS

MultiSenseNet: A Unified Approach to Multi-Task Machine Learning | [Poster](#)

Advancing Machine Olfaction for Real-World Allergen Detection | [Poster](#)

AI4Climate: Collection of ML tasks and Datasets for Tackling Climate Change | [Poster](#)

HONORS AND AWARDS

MIT Break Through Tech AI Scholar

August 2025

- Selected for MIT's applied AI program focused on data science, ethics, and real-world machine learning through mentorship and hands-on projects.

Stanford Deep Learning Portal Program Scholar

April 2024

- Selected for Stanford's Deep Learning Portal Program, a competitive initiative offering mentorship and advanced training in deep learning and AI research.

Tau Beta Pi Engineering Honor Society Scholar

March 2024

- Inducted into the nation's oldest engineering honor society for academic excellence, integrity, and leadership in engineering.

Howard University Hackathon - 1st Place Winner

March 2024

- Won 1st place at Howard's Bison Hacks for developing MindMatters, an AI chatbot assisting first-generation students with academic, financial, and career guidance.

Distinction Award, BCF Leadership Training Institute

August 2023

- Earned distinction upon completing a leadership program focused on mentorship, integrity, and service, emphasizing character and community leadership.

NSTMF Instem Scholar

September 2022

- Selected for the National Science & Technology Medals Foundation mentorship program supporting underrepresented students in STEM.

Achievers Scholarship

August 2022

- Merit-based scholarship awarded to top-performing students; covers a portion of tuition and recognizes strong academic achievement.

NBCUniversal Scholarship

Spring 2023 – Fall 2024

- Merit-based award from NBCUniversal supporting academic excellence and career development through tuition assistance.

Dean's List

2022 – Present

- Recognized for maintaining a high GPA and academic excellence, reflecting consistent performance and dedication throughout my undergraduate studies at Howard University.

PROFESSIONAL AFFILIATIONS

Bethel Campus Fellowship

Fall 2023 – Present

- Engage in faith-based community service and leadership activities that promote character and personal growth.

Howard University CEA Student Council

Spring 2022 – Spring 2024

- Collaborated on student government initiatives to improve academic and community engagement within the College of Engineering and Architecture.

Institute of Electrical and Electronics Engineers (IEEE)

August 2022 – Present

- Member of IEEE, participating in professional development, technical workshops, and networking events with engineering peers.

National Society of Black Engineers (NSBE)

August 2022 – Present

- Member of NSBE, engaging in activities fostering excellence and leadership in engineering.

NSTMF InSTEM Scholars Program

August 2022 – Present

- Participating in a program aimed at supporting STEM students through mentorship and resources.

CONFERENCES PARTICIPATION

Bethel Campus Fellowship National Conference

February 2025

- Participated in workshops and leadership sessions focused on faith, purpose, and community impact among students nationwide.

NSBE's 50th Annual Convention

March 2024

- Engaged in career development workshops and networking events promoting engineering excellence and diversity in STEM.

Amazon Tech Career Day

March 2024

- Attended tech talks, leadership and resume workshops, AWS Skills Center tour, and mock interviews with Amazon engineers to gain career and industry insights.

TMCF DevCon

June 2023

- Attended the Thurgood Marshall College Fund's Developer Conference focused on technology, innovation, and career advancement through workshops, panels, and networking with industry professionals.

COMMUNITY OUTREACH AND SERVICE

STEM Outreach Initiative

June 2025

- Supported a technology fair hosted by Howard University and the District Department of Transportation, engaging high school students in interactive engineering demonstrations to spark interest in STEM careers.

Shoe Drive Outreach

May 2025

- Organized a shoe drive on campus in collaboration with the Nigerian Students Organization, collecting and donating gently used footwear to underprivileged children in Nigeria.

Howard University Alternative Spring Break

March 2023

- Assisted with home restoration in underserved communities in Puerto Rico impacted by natural disasters, helping to revitalize living spaces and uplift residents' quality of life through service and teamwork.

REFERENCES

Prof. Priya Donti – Assistant Professor, MIT EECS & LIDS

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Prof. Paul Pu Liang – Assistant Professor, MIT Media Lab & EECS

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Prof. Jeremy Blackstone – Assistant Professor, Howard University EECS

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